

455 Random assembled notes

Nowak Nowak

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1 Purpose

This document collects the key calculations used to (1) estimate the polarization coil electrical parameters and (2) size the switching transistors and protection network for a 12V battery-powered PPM pulser.

2 Known coil geometry (polarization coil)

From the coil design:

- Inner diameter: 11.4cm (4.488in) $\Rightarrow r_{in} = 5.7cm$
- Coil length: 10.0cm (3.937in) $\Rightarrow \ell = 0.10m$
- Turns: $N = 228$ (6 layers $\times$ 38 turns/layer)
- Wire: 14 AWG copper (bare diameter $\approx 1.628mm$; insulated diameter depends on enamel/build)

For multilayer coils, the inductance depends on the **mean radius** and **winding thickness**: $r_{avg} \approx r_{in} + \frac{t}{2}$
 $t \approx N_{layers} \cdot d_{wire,ins}$ A reasonable first estimate is $d_{wire,ins} \sim 1.7mm$, giving $t \approx 6 \times 1.7 \approx 10.2mm$ and $r_{avg} \approx 6.2cm$.

3 Inductance estimate

A standard engineering estimate for a multilayer solenoid is the Wheeler formula:

$$L(\mu H) \approx \frac{0.8 r^2 N^2}{6r + 9\ell + 10t} \quad (1)$$

where r , ℓ , and t are in **inches** (mean radius, coil length, winding thickness).

Using:

$$r \approx 2.44in, \quad \ell \approx 3.94in, \quad t \approx 0.40in, \quad N = 228$$

$$L \approx \frac{0.8(2.44)^2(228)^2}{6(2.44) + 9(3.94) + 10(0.40)} \mu H \approx 4.6e3H \approx 4.6mH$$

Design value: $L \approx 4.6mH$ (*air – core estimate*).

Uncertainty

Real inductance can vary $\pm 10\%$ to $\pm 25\%$ due to winding packing, insulation thickness, nearby conductors/metal, and measurement frequency. For design, it is honest to state:

$$L \approx 46mH \text{ expected; measure to confirm.}$$

4 Wire length and DC resistance estimate

Approximate total wire length using mean circumference: mean circumference

$$C \approx 2\pi r_{avg}$$

$$L_{wire} \approx N \cdot C \text{ With } r_{avg} \approx 0.062m:$$

$$C \approx 2\pi(0.062) \approx 0.389m, \quad L_{wire} \approx 228(0.389) \approx 88.7m$$

Copper resistance for 14 AWG is approximately $0.00829m$ at $20C$.

$$R_{coil} \approx 0.00829 \times 88.7 \approx 0.74$$

Design value: $R_{coil} \approx 0.7$ (*estimate; measure*)

Temperature rise

Copper increases with temperature:

$$R(T) = R_{20} [1 + \alpha(T - 20^\circ C)], \quad \alpha \approx 0.00393/^\circ C \quad (2)$$

Example: at $60C$, $R \approx R_{20}(1 + 0.00393 \cdot 40) \approx 1.16R_{20}$.

5 Pulse current and polarization field

5.1 Peak current (battery-limited)

For a 12V battery and low-side MOSFET switch, the peak current is approximately:

$$I_{peak} \approx \frac{V_{bat}}{R_{coil} + R_{switch} + R_{wiring}} \quad (3)$$

Ignoring small parasitics for first-pass:

$$I_{peak} \approx \frac{12}{0.74} \approx 16A$$

(*This is why low $R_{DS(on)}$ and thick wiring matter.*)

5.2 Field at center (finite solenoid correction)

For a finite-length solenoid, the on-axis field at the center is:

$$B(0) \approx \mu_0 \frac{N}{\ell} I \cdot \frac{\ell}{2\sqrt{r^2 + (\ell/2)^2}} = \mu_0 \frac{NI}{2\sqrt{r^2 + (\ell/2)^2}} \quad (4)$$

Using $r \approx 0.062m$, $\ell = 0.10m$, $N = 228$, $I = 16A$:

$$\frac{1}{2\sqrt{r^2 + (\ell/2)^2}} \approx \frac{1}{2\sqrt{0.062^2 + 0.05^2}} \approx 6.28$$

$$B(0) \approx \mu_0 NI(6.28) \approx (1.256 \times 10^{-6})(228)(16)(6.28) \approx 2.9e-2T \approx 29mT$$

Interpretation: tens of mT is a strong polarization field for a DIY PPM head (actual value depends on geometry and end effects).

6 Inductor energy and turn-off stress (why clamps matter)

The energy stored in the polarization coil at the end of the pulse is:

$$E_L = \frac{1}{2}LI^2 \quad (5)$$

With $L = 4.6mH$ and $I = 16A$:

$$E_L \approx 0.5(0.0046)(16^2) \approx 0.59J$$

This energy must go somewhere at turn-off: into a TVS, a snubber, MOSFET avalanche, or back into the supply (active clamp).

6.1 Current decay rate vs clamp voltage

For low-side switching with coil top at +12 V and switch node clamped to V_{clamp} :

$$V_L = V_{top} - V_{bottom} = 12 - V_{clamp}$$

$$\frac{dI}{dt} \approx \frac{V_L}{L} = -\frac{(V_{clamp} - 12)}{L}$$

Example:

- **Diode-only clamp to +12 V** (effectively $V_{clamp} \approx 12.7$):
 $|V_L| \approx 0.7V \Rightarrow dI/dt \approx -0.7/0.0046 \approx -150A/s$
 $\Rightarrow 16A$ takes $\sim 0.10s$ to decay (slow).
- **TVS clamp to 60V** ($V_{clamp} = 60$):
 $|V_L| \approx 48V \Rightarrow dI/dt \approx -48/0.0046 \approx -1.0e4A/s$
 $\Rightarrow 16A$ decays in $\sim 1.5ms$ (fast).

PPM implication: faster decay generally means shorter receiver blanking time and faster recovery.

7 MOSFET sizing and key checks

7.1 Voltage rating

MOSFET V_{DS} must exceed the worst-case switch node excursion:

$$V_{DS,max} V_{clamp,peak} + V_{ringing} + margin$$

Practical guidance:

- If using a TVS clamp in the 5080V region, **100 V MOSFETs can work** but margin is tight.
- **150 V MOSFETs are safer** for early prototypes and unknown ringing.

7.2 Conduction loss (during pulse)

For n MOSFETs in parallel:

$$R_{eq} \approx \frac{R_{DS(on)}}{n}, \quad P_{cond} \approx I^2 R_{eq}$$

Example: if each MOSFET is 6m and $n = 3$, $R_{eq} \approx 2m$:

$$P_{cond} \approx 16^2(0.002) \approx 0.51W \text{ (during pulse)}$$

Average power is reduced by duty cycle.

7.3 Gate drive requirement

Total gate charge adds in parallel:

$$Q_{total} \approx nQ_g, \quad I_{gate} \approx \frac{Q_{total}}{t_{sw}}$$

Example: $Q_g = 120nC$ each, $n = 3 \Rightarrow Q_{total} = 360nC$. To switch in $t_{sw} = 500ns$:

$$I_{gate} \approx \frac{360nC}{500ns} \approx 0.72A$$

Implication: a dedicated gate driver (multiple amps peak) is recommended for clean switching.

7.4 Per-device gate resistors

When paralleling MOSFETs, use one resistor per gate (typ. 1022) to reduce ringing and improve sharing.

8 Clamp / snubber energy checks (TVS sizing intuition)

Worst-case, assume the TVS absorbs most of E_L each turn-off:

$$E_{pulse} \approx E_L \approx 0.6J$$

If the cycle period is T_{cycle} :

$$P_{avg} \approx \frac{E_{pulse}}{T_{cycle}}$$

Example: one pulse every 5s $\Rightarrow P_{avg} \approx 0.6/5 = 0.12W$ (average is small), but the TVS must survive the **pulse** current/energy.

9 Battery draw and runtime (back-of-envelope)

If pulse current is I_{peak} , pulse width t_p , and cycle period T :

$$I_{avg} \approx I_{peak} \frac{t_p}{T}$$

Example: $I_{peak} = 16A$, $t_p = 1s$, $T = 5s$:

$$I_{avg} \approx 16 \cdot \frac{1}{5} = 3.2A$$

A 12V, 9Ah battery ideally provides:

$$t_{ideal} \approx \frac{9}{3.2} \approx 2.8h$$

Real runtime will be lower due to voltage sag, temperature, and Peukert effects.

10 How to measure L without an LCR meter (recommended)

Put a known power resistor R_s in series with the coil and apply a step (low duty). Measure current rise via the resistor:

$$i(t) = I_{\infty} \left(1 - e^{-t/\tau}\right), \quad \tau = \frac{L}{R_s + R_{coil}}$$

Pick R_s so the current is safe. Fit τ from the waveform and compute:

$$L = \tau (R_s + R_{coil})$$

11 Summary table (design starting numbers)

Quantity	Estimate	Notes
Inductance L	4.6mH	Wheeler multilayer estimate
Resistance R_{coil}	0.7	from wire length
Peak current @ 12V	16A	neglecting switch/wiring drops
Stored energy E_L	0.6J	$0.5LI^2$
Center field (rough)	30mT	finite-solenoid correction